```bash echo -e "Hello, I'm a language model" | transformers run --task text-generation --model openai-community/gpt2 --device 0 ```

https://huggingface.co/docs/transformers/main/en/model_doc/gpt2#transformers

Detailed Documentation

class <span class="code-inline">transformers.GPT2Configtransformers.GPT2Confighttps<

class `transformers.GPT2Configtransformers.GPT2Confighttps` ://github.com/huggir "vocab_size", "val": " = 50257"}, {"name": "n_positions", "val": " = 1024"}, {"name": "n_embd", "val": " = 768"}, {"name": "n_layer", "val": " = 12"}, {"name": "n_head", "val": " = 12"}, {"name": "n_inner", "val": " = None"}, {"name": "activation_function", "val": " = 'gelu_new'"}, {"name": "resid_pdrop", "val": " = 0.1"}, {"name": "embd_pdrop", "val": " = 0.1"}, {"name": "attn_pdrop", "val": " = 0.1"}, {"name": "layer_norm_epsilon", "val": " = 1e-05"}, {"name": "initializer_range", "val": " = 0.02"}, {"name": "summary_type", "val": " = 'cls_index'"}, {"name": "summary_use_proj", "val": " = True"}, {"name":
```

"summary_activation", "val": " = None"}, {"name": "summary_proj_to_labels", "val": " = True"}, {"name": "summary_first_dropout", "val": " = 0.1"}, {"name": "scale_attn_weights", "val": " = True"}, {"name": "use_cache", "val": " = True"}, {"name": "bos_token_id", "val": " = 50256"}, {"name": "eos_token_id", "val": " = 50256"}, {"name": "scale_attn_by_inverse_layer_idx", "val": " = False"}, {"name": "reorder_and_upcast_attn", "val": " = False"}, {"name": "**kwargs", "val": ""}]-
vocab_size (`int`, *optional*, defaults to 50257) --

Vocabulary size of the GPT-2 model. Defines the number of different tokens that can be represented by the

`inputs_ids` passed when calling [GPT2Model](/docs/transformers/main/en/model_doc/gpt2#`transformers.GPT2Model`) or `TFGPT2Model`.

- **n_positions** (`int`, *optional*, defaults to 1024) --

The maximum sequence length that this model might ever be used with. Typically set this to something large

just in `case` (e.g., 512 or 1024 or 2048).

- **n_embd** (`int`, *optional*, defaults to 768) --

Dimensionality of the embeddings and hidden states.

- **n_layer** (`int`, *optional*, defaults to 12) --

Number of hidden layers in the Transformer encoder.

class <span class="code-inline">transformers.GPT2Tokenizertransformers.GPT2Tokenizer

class `transformers.GPT2Tokenizertransformers.GPT2Tokenizerhttps`://github.com/ "vocab_file", "val": ""}, {"name": "merges_file", "val": ""}, {"name": "errors", "val": " = 'replace'"}, {"name": "unk_token", "val": " = '<|endoftext|>'"}, {"name": "bos_token", "val": " = '<|endoftext|>'"}, {"name": "eos_token", "val": " = '<|endoftext|>'"}, {"name": "pad_token", "val": " = None"}, {"name": "add_prefix_space", "val": " = False"}, {"name": "add_bos_token", "val": " = False"}, {"name": "**kwargs", "val": ""}]- **vocab_file** (`str`) --

Path to the vocabulary file.

- **merges_file** (`str`) --

Path to the merges file.

- **errors** (`str`, *optional*, defaults to `"replace"`) --

Paradigm to follow when decoding bytes to UTF-8. See

[bytes.decode](https://docs.python.org/3/library/stdtypes.html#bytes.decode) for more information.

- **unk_token** (`str`, *optional*, defaults to `"<|endoftext|>"`) --

The unknown token. A token that is not in the vocabulary cannot be converted to an ID and is set to be this

- **bos_token** (`str`, *optional*, defaults to `"<|endoftext|>"`) --

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save_vocabularytransformers.GPT2Tokenizer.save_vocabularyht

save_vocabularytransformers.GPT2Tokenizer.save_vocabularyhttps://github.com/ "save_directory", "val": ": str"}, {"name": "filename_prefix", "val": ": typing.Optional[str] = None"}]

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class <span class="code-inline">transformers.GPT2TokenizerFasttransformers.GPT2Toker

class `transformers.GPT2TokenizerFasttransformers.GPT2TokenizerFasthttps`://git "vocab_file", "val": " = None"}, {"name": "merges_file", "val": " = None"}, {"name": "tokenizer_file", "val": " = None"}, {"name": "unk_token", "val": " = '<|endoftext|>'"}, {"name": "bos_token", "val": " = '<|endoftext|>'"}, {"name": "eos_token", "val": " = '<|endoftext|>'"}, {"name": "add_prefix_space", "val": " = False"}, {"name": "**kwargs", "val": ""}]- **vocab_file** (`str`, *optional*) --

Path to the vocabulary file.

- **merges_file** (`str`, *optional*) --

Path to the merges file.

- **tokenizer_file** (`str`, *optional*) --

Path to [tokenizers](https://github.com/huggingface/tokenizers) `file` (generally has a .json extension) that

contains everything needed to load the tokenizer.

- **unk_token** (`str`, *optional*, defaults to `"<|endoftext|>"`) --

The unknown token. A token that is not in the vocabulary cannot be converted to an ID and is set to be this

- **bos_token** (`str`, *optional*, defaults to `"<|endoftext|>"`) --

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class <span class="code-inline">transformers.models.gpt2.modeling_gpt2.GPT2DoubleHe

class `transformers.models.gpt2.modeling_gpt2.GPT2DoubleHeadsModelOutputtrans` "loss", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "mc_loss", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "logits", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "mc_logits", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "past_key_values", "val": ": typing.Optional[`transformers.cache_utils.Cache`] = None"}, {"name": "hidden_states", "val": ": typing.Optional[tuple[torch.FloatTensor]] = None"}, {"name": "attentions", "val": ": typing.Optional[tuple[torch.FloatTensor]] = None"}]- **loss** (`torch.FloatTensor` of shape `(1,)`, *optional*, returned when `labels` is provided) --

Language modeling loss.

- **mc_loss** (`torch.FloatTensor` of shape `(1,)`, *optional*, returned when `mc_labels` is provided) --

Multiple choice classification loss.

- **logits** (`torch.FloatTensor` of shape `(batch_size, num_choices, sequence_length, config.vocab_size)`) --

Prediction scores of the language modeling `head` (scores for each vocabulary token before SoftMax).

- **mc_logits** (`torch.FloatTensor` of shape `(batch_size, num_choices)`) --

Prediction scores of the multiple choice classification `head` (scores for each choice before SoftMax).

- **past_key_values** (`Cache` , *optional*, returned when `use_cache=True` is passed or when `config.use_cache=True`) --

It is a [Cache](/docs/transformers/main/en/internal/generation_utils# `transformers.Cache`) instance. For more details, see our [kv cache guide](https://huggingface.co/docs/transformers/en/kv_cache).

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class <span class="code-inline">transformers.GPT2Modeltransformers.GPT2Modelhttps</

class

`transformers.GPT2Modeltransformers.GPT2Modelhttps`://github.com/huggingf "config", "val": ""}]- **config** ([GPT2Model] (/docs/transformers/main/en/model_doc/gpt2#`transformers.GPT2Model`)) --

Model configuration class with all the parameters of the model. Initializing with a config file does not

load the weights associated with the model, only the configuration. Check out the

[`from_pretrained`()] (/docs/transformers/main/en/main_classes/model#`transformers.PreTrainedMoc` method to load the model weights.0

The bare Gpt2 Model outputting raw hidden-states without any specific head on top.

This model inherits from [PreTrainedModel] (/docs/transformers/main/en/main_classes/model#`transformers.PreTrainedMoc` Check the superclass documentation for the generic methods the

library implements for all its `model` (such as downloading or saving, resizing the input embeddings, pruning heads

This model is also a PyTorch [torch.nn.Module] (https://pytorch.org/docs/stable/nn.html#torch.nn.Module) subclass.

Use it as a regular PyTorch Module and refer to the PyTorch documentation for all matter related to general usage forwardtransformers.GPT2Model.forwardhttps://github.com/huggingface/transfor "input_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "past_key_values", "val": ": typing.Optional[`transformers.cache_utils.Cache`] = None"}, {"name": "cache_position", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "attention_mask", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "token_type_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "position_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "inputs_embeds", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "encoder_hidden_states", "val": ": typing.Optional[torch.Tensor] = None"}, {"name": "encoder_attention_mask", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "use_cache", "val": ": typing.Optional[bool] = None"}, {"name": "output_attentions", "val": ": typing.Optional[bool] = None"}, {"name": "output_hidden_states", "val": ": typing.Optional[bool] = None"}, {"name": "return_dict", "val": ": typing.Optional[bool] = None"}, {"name": "**kwargs", "val": ""}]- **input_ids** (`torch.LongTensor` of shape `(batch_size, input_ids_length)`) --

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forwardtransformers.GPT2Model.forwardhttps://github.com/hugg

forwardtransformers.GPT2Model.forwardhttps://github.com/huggingface/transfor "input_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "past_key_values", "val": ": typing.Optional[`transformers.cache_utils.Cache`] = None"}, {"name": "cache_position", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "attention_mask", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "token_type_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "position_ids", "val": ": typing.Optional[torch.LongTensor] = None"}, {"name": "inputs_embeds", "val": ": typing.Optional[torch.FloatTensor] =

None"}, {"name": "encoder_hidden_states", "val": ": typing.Optional[torch.Tensor] = None"}, {"name": "encoder_attention_mask", "val": ": typing.Optional[torch.FloatTensor] = None"}, {"name": "use_cache", "val": ": typing.Optional[bool] = None"}, {"name": "output_attentions", "val": ": typing.Optional[bool] = None"}, {"name": "output_hidden_states", "val": ": typing.Optional[bool] = None"}, {"name": "return_dict", "val": ": typing.Optional[bool] = None"}, {"name": "**kwargs", "val": ""}]- **input_ids** (`torch.LongTensor` of shape `(batch_size, input_ids_length)`) --

`input_ids_length` = `sequence_length` if `past_key_values` is `None` else

`past_key_values.` `get_seq_length` `()` (`sequence_length` of input past key value states). Indices of input

sequence tokens in the vocabulary.

If `past_key_values` is used, only `input_ids` that do not have their past calculated should be passed as

Indices can be obtained using [AutoTokenizer] (/docs/transformers/main/en/model_doc/auto# `transformers.AutoTokenizer`). See [PreTrainedTokenizer. `encode` ()] (/docs/transformers/main/en/internal/tokenization_utils# `transformers.PreTraine`
and

[PreTrainedTokenizer. `__call__` ()] (/docs/transformers/main/en/internal/tokenization_utils# `transformers.PreTraine`
for details.

[What are input IDs?](../glossary#input-ids)

- **past_key_values** (`~cache_utils.Cache`, *optional*) --

Pre-computed hidden-`states`(key and values in the self-attention blocks and in the cross-attention

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class <span class="code-inline">transformers.GPT2LMHeadModeltransformers.GPT2LMH
